Léo Kakali da Fonseca Souza

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Summary

Informatics Engineering student at the University of the Algarve (17/20), graduating in June 2027 and available from July 2027. Backend developer in Python (FastAPI, SQLAlchemy, pytest), working with code review, automated testing and CI/CD on GitHub Actions. I use AI-assisted tools (Claude Code) daily, measure performance with benchmarks, and want to apply Generative AI across the software development life cycle.

Skills

Programming Languages
Python, C++, Java, SQL, OCaml (Functional Programming), Typescript/JavaScript

Systems & Backend
PostgreSQL, REST APIs, FastAPI, JWT, algorithms and complexity, benchmarking, cache locality, HMAC signing

Infrastructure
Docker, Docker Compose, Linux, GitHub Actions CI/CD, Alembic migrations

Languages
English (Professional working), Portuguese (Native), Spanish (Basic)

Secure & AI-assisted development
PII-free data design, Claude Code for code generation, refactoring and tests, validated through review, automated tests and CI

Engineering Practice
TDD, JUnit 5, pytest, Git, Maven

Experience

ENEI 2027 | Website — Faro, Portugal
Volunteer Backend Developer — Jun 2026 – Present

- Every change goes through CI/CD on GitHub Actions (ruff, pytest on PostgreSQL, smoke test) before deploy; Docker Compose stack of 8 services.
- Built the authentication and accounts flows of the ticketing API (FastAPI, SQLAlchemy 2, PostgreSQL): JWT auth, password recovery, terms acceptance and an admin CLI, across 12 merged PRs.
- Reviewed 11 PRs from 3 teammates; wrote over 25 of the 500+ tests and 4 of the 20 Alembic migrations, validated in CI as reversible.
- Designed an HMAC-signed ticket QR payload with no personal data, so tickets can be verified without exposing attendee information.

Education

University of Algarve — BSc Informatics Engineering • grade 17/20
Portugal • Sep 2024 – Jun 2027

- Grade 17/20 · Algorithms and Data Structures 20/20 · Object-Oriented Programming 20/20 · Software Engineering 18/20.
- SOUAlg2.0 Academic Merit Scholarship · Member of GT TRUST-Lab (OCaml, arbitrary-precision arithmetic).

Projects

High-Performance Data Structures & Memory Benchmark (FintList) | Github

- Doubly linked list built on an integer array, with slot reuse and caching of the last access. Maven, JUnit 5, GitHub Actions.
- About 8× faster than the course's linked list on addAt/removeAt with ~10 million elements, measured with doubling-ratio benchmarks; both are O(n), the gain comes from smaller constants.

Maritime Navigation & Route Planning Simulator | Github
MVC separation between the simulation engine and a Swing interface; Strategy (Dijkstra route planning that discards segments crossing obstacles), State and Memento patterns; JUnit 5 and Maven build run in GitHub Actions.

Awards

Competitive Programming (ICPC / SWERC & MIUP) | ICPC ID

- MIUP 2026 (Évora): solved 4 problems in a 3-person team, up from 0 at MIUP 2025 (Aveiro), after a year of training on graph, DP and data-structure problems.
- SWERC 2025 (Lisbon): represented UAlg at the ICPC regional; I maintain the team's C++ reference notebook: github.com/kakurokan/SwercNotebookUALG